Embroiderer: Do-It-Yourself Embroidery Aided with Digital Tools

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ABSTRACT

Traditional crafts such as knitting, embroidery and weaving are becoming increasingly popular due to their low cost and high accessibility, creating products with unique value. However, the lack of practical experience and the complexity of the handicraft often limit the participation of handicraft enthusiasts. In this paper, we develop a computer-aided embroidery method, called Embroiderer, which combines design tools to help novice users convert digital images into real embroidery processes. In detail, users can customize, modify and preview their designs in Embroiderer with knitting threads of different colors and thicknesses; the desired designs can be served as craft-aided molds which can be laser-cut or 3D printed with water-soluble materials; thus, users can easily embroider patterns on fabric following instructions from nodes embedded in the molds. Through a series of experiments and design cases, we hope to provide an accessible approach to the hybrid craft process in the pattern transformation of everyday clothing, fabrics or textiles.

CCS CONCEPTS •Human-centered computing •Human computer interaction (HCI) •Computational Tool

KEYWORDS

Handicraft, Personalization, Computer-aided Craft Design, Digital Craft

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Figure 1: Overview of Embroiderer: Users can input (a) any picture in our design tool, (b, c) generate mold vector path that conforms to the pattern, which can be 3D printed or laser cut through digital manufacturing tools to generate knitting molds; The user can (d) then easily thread, (e) the process-assisted mold embroidery pattern that can be dissolved in water; Thus, users can (f, g) get their own DIY clothes, fabrics or textiles.

1 INTRODUCTION

Embroidery and weaving crafts are among the oldest traditional human crafts, which used to aid survival during the Paleolithic period about 3 million years ago according to the Book of Changes (I Ching) [9]. It has also been a popular craft practice over the years, not only can it produce a wide variety of woven products with diverse patterns, but it also helps to improve many abilities, for example, promoting hand-eye coordination, improving thinking skills, and even stimulating the development of motor skills in children [4]. However, traditional embroidery technology usually requires much practice time and fine skills, which is often a high threshold for beginners. At the same time, due to the development of the social economy and the modern aesthetic, people have more personalized style requirements for daily products [2].

In recent years, with the development of DIY technology, such as digital design and fabrication tools, the Human-computer interaction (HCI) community has increased to leverage digital manufacturing and develop assistive tools to create "easier to make and more diverse" handicrafts. For example, Amit proposed "Hybrid craft" to showcase a series of case studies of physical and digital integration of design and craft skills [3, 6, 8, 13]; Liu, et al [7] explore digital making as an approach for the inheritance of HCI handicrafts. In the weaving craft, Ada-CAD supports the simultaneous design of woven structures and circuits, helping to overcome some of the challenges associated with designing woven circuits [10]; Escape provides several weaving techniques enabled by 3D printed custom tools to afford new weaving experiences [1]; Punch needle embroidery [5] provides a unique type of embroidery that uses loops of threads to create designs. While there is a growing body of research utilizing technology to aid embroidery, relatively little attention has been paid to utilizing computing technology to facilitate the embroidery process and incorporate personalized design methods.

In this paper, we try to combine computing methods within traditional processes and generate patterns that help novice users design and craft through a parametric design tool. To achieve this vision, we conducted a series of experiments on the material of aided mold and the thicknesses of the knitting thread. The results are adopted in developing a personalized design tool that can benefit the automatic generation of patterns for handicrafts. In the crafting process, users can use 3D printing or laser cutting to fabricate craft-aided molds, which can help simplify the embroidery process for novice users [Figure 1] . Our method can be applied to custom fabrics, textiles, or clothes, which has great potential in hybrid embroidery in personal fabrication.

The contributions of this paper include:

Proposing a hybrid embroidery workflow to combine digital technology with traditional embroidery craft, exploring ways to lower the threshold of manual embroidery.

Developing a design tool to generate the craft aided molds, with the exploration of the materials used in the mold and the optimized parameters of punching schemes.

Conducting a user study to verify the feasibility of this method and providing insight findings to explore the potential of hybrid embroidery.

2 RELATED WORKS AND BACKGROUND

2.1 Hybrid Craft in HCI

The field of hybrid craft is experiencing robust growth and has garnered significant attention in HCI. In practical applications, digital fabrication tools delineate pathways for users to obtain correct solutions 12 , expanding the scope of individual manufacturing while reducing the time and complexity involved in creative processes. For example, CompuWoven [14] demonstrates a system that pertains to innovative manual weaving methods, facilitating the customization and weaving of 3D objects from planar blocks. WeaveMesh [15] develops a low-fidelity, cost-effective rapid prototyping system capable of generating 3D objects within grid structures. 4Doodle [17] proposes a hybrid craft approach that integrates unique deformability and five manual 4D printing techniques. PneuFab [11] proposes a novel and democratized blow molding technique that enabled by FDM 3D-printing technology.

2.2 Computer-Aided Traditional Handicraft

In niche domains, numerous endeavors employ digital craftsmanship to preserve traditional handicraft. For instance, Tao et al. [13] developed an interactive system to support the Bamboo Craft-Design process, from design inspiration to pattern abstraction and product creation. Jacobs et al. [5] integrated digital tools with the traditional ostrich eggshell jewelry craft of the South African people in the Kalahari. InfOrigami [16] employs computer-aided design to manufacture traditional perforated boneless lantern crafts. Inspired by these efforts, we have created a digital fabrication tool with the aim of lowering the skill barrier associated with hardware and software use, and increasing the motivation for new users to experiment.

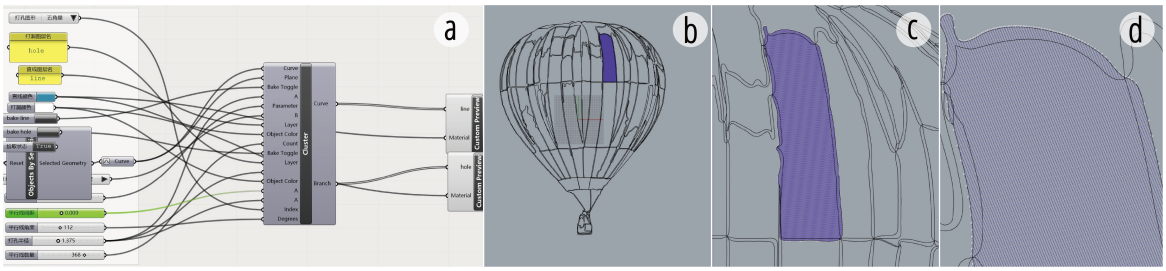


Figure 2: The design tool was developed with (a) the parametric tools to design and preview (b-d) the results of line arrangement in real-time.

2.3 Weaving Craft and DIY Toolkits

Textiles represent an ancient and deeply intertwined facet of human life. Many scholars have dedicated themselves to the study and advancement of textiles, attempting to integrate them with the latest technologies and explore their broader potential. For example, the E-Sewing Tutorial [18] provides an in-depth guide on how to sew electronic circuits into fabric. Toolkit furnishes a generative and open-ended creative space within the solution domain, enabling users to construct different solutions by reusing, combining, and adjusting the provided building blocks [20]. Related works in the realm of weaving include E-broidery [19], which uses multi-layer electronic circuits as the foundation of textile work. Ebb [21] applies conductive thread coated with thermochromic pigment, resulting in various woven and crochet fabric color palettes. E-Textiles alternate between embroidery and sewing, incorporating real electrical components into the fabric. EscapeLoom [1] introduces a novel weaving technique involving water-soluble sketches, flexible warp threads, and rigid global geometries. In this paper, we will explore the additional possibilities of textile toolkits, aiming to provide information for the future development of textiles by enhancing user engagement and exploration 22.

3 EMBROIDERER SYSTEM

3.1 Design Concept

Regarding the background of the bridge between traditional embroidery and the DIY demand from people, we developed a computer-aided design and craft method to democratize a professional embroidery process for novice users, who may have difficulty in traditional pattern design and embroidery techniques.

Based on the literature research [3, 6, 8, 9, 13] and field observation, we investigated how to customize the patterns of traditional embroidery, such as color and line size, in graphic design insight. Taking into account potential DIY elements in the embroidery, we proposed a design concept to customize the arrangement of holes of a mold for the guidance of the embroidery process. In addition, advanced personal fabrication tools, such as 3D printing technology, allow users to access them directly to fabricate craft-aided mold.

3.2 Overview

In our Embroiderer system, we develop a design tool based on Rhino and Grasshopper plugins, which can generate the craft-aided mold by outlining the imported patterns, embedding holes with different spacing, shapes, and position in the mold, and preview the results of line arrangement in real-time (Figure 2).

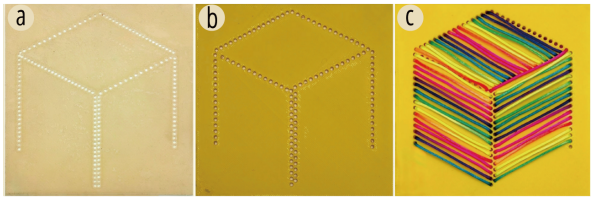


Figure 3: Material test of the molds: the molds can be 3D printed with (a) PVA, (b) PLA, which can be woven with knitting thread (c).

Based on the prepared cotton threads, users can first design and automatically generate the mold in the design tool according to an identified pattern with different colors and sizes. Users can determine expected product size and cotton thread diameter, punch holes for the model, adjust hole sizes and parallel line spacing, and print the model using PVA using a consumer-grade 3D printer. After these, users can use plastic needles to aid the embroidery process, as it is safe and easy to thread, and complete the pattern according to the distribution of holes in the tool. Finally, users can put the finished product into the water to completely dissolve the mold, thus getting the final embroidery work.

4 EXPERIMENTS

To better improve the accuracy and feasibility of Embroiderer, we conduct a series of experiments, including the mold material, the hole size of the mold and the spacing between the braided thread.

4.1 Material Selection

Leveraging the low-cost advantage of fused deposition modeling (FDM) 3D printing technology, we use standard 3D printing filaments as experimental materials, such as PLA, ABS, and PVA, which are biodegradable materials. (Figure 3) It is worth noting that PVA is highly soluble and can dissolve quickly in warm water, reducing the amount of post-processing required for printouts and help reduce the risk of damaging the work.

4.2 Diameters of Holes

For braided wires of different diameters, it's crucial to determine the appropriate hole size in our tool. We conducted extensive tests to measure the hole sizes needed for different braided thread diameters. Considering the effect of thread density on the embroidery effect, we weighed the errors in the 3D printing process and the hardness of the hole-to-hole connection to conclude a series of optimized parameters.

Building on previous discussion, we selected the appropriate woven wire materials and 3D printing technology, and a total of 3 different thicknesses of woven materials were used to test the diameter of the holes. As shown in Figure 4, our measurement data are mainly carried out around three thickness lines of 1.5mm, 2.0mm and 2.5mm. Through a series of experiments, the final result is that the 1.5mm thick yarn is suitable for the 2.75mm aperture, and the 2.0mm and 2.5mm yarn is matched with the 3.115mm and 3.625mm aperture respectively, so that the final visual effect can be achieved.

4.3 The Thickness and Spacing of Thread

In order to better meet the different needs of users, we need to study the most suitable thickness and spacing calculation of knitting thread. In the wiring process, the thickness of the wire directly affects the shape of the wire. We simulated the presentation of different shapes, taking into account the thickness and shape of lines required for holes of different sizes. By calculating the line spacing, we can ensure that during the wiring process, the size of the line fits perfectly with the path, which can clearly display the user's design.

In the line space test, we tried several molds with different hole gaps. For example, for the 1.5mm thickness wire, we test the three types of hole spacing of 0.05mm, 0.12mm, and 0.22mm. Due to the small corresponding cavity diameter, it is difficult to control the printing accuracy, and there are errors in embroidery. Applicable to the remaining two kinds of wires, the result shows that the most suitable data are respectively 2.75mm diameter hole, 0.15mm parallel line spacing for 1.5mm thickness wire, and 3.28mm diameter hole, 0.25mm parallel line spacing for 2.5mm thickness wire.

5 DESIGN TOOL

Our design tool, Embroiderer, provides a de-professionalizing method for making the process of knitting highly accessible for novice users, who may struggle with traditional pattern design and embroidery techniques due to inadequate knowledge. Benefiting from pattern-assisted design tools and 3D printing molds, our design tool can help users easily create personalized patterns, which aims to develop a computational system based on the Rhino and Grasshopper plug-ins by customizing the imported patterns to generate a mold in which parametric holes of different pitches, shapes, and positions, thus can assist in hand embroidery the corresponding patterns and previewing the real-time embroidery results (Figure 5).

5.1 Import of the Custom Pattern Image

In conventional design methodologies, users typically adhere to predefined embroidery patterns set by the designer. In contrast, our tool provides users the autonomy to choose a bespoke pattern (Figure 5a).

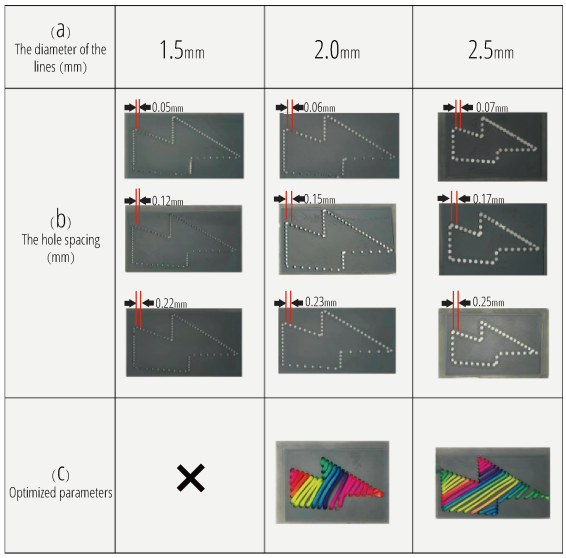


Figure 4: The optimized parameters of mold design based on a series of experiments.

5.2 Automatic Generation of Vectorized Pattern Borders

Our tool transcends merely simplifying the skill requirements; it also streamlines the traditionally intricate process of embroidery pattern design, transitioning from image abstraction to actual embroidery patterns. Furthermore, it intelligently delineates embroidery regions based on color distribution (Figure 5b).

5.3 Weave Path Settings

The tool offers users the capability to dictate the direction of the weaving path in alignment with the pattern (Figure 5c). This approach not only fosters a more methodical embroidery outcome but also bolsters individual customization.

5.4 Auxiliary Embroidery Nodes Settings

Through customized design of nodes settings, the difficulty of embroidery manual work can be effectively reduced and the design space of embroidery work can be expanded.

Pitch setting: The purpose is to provide different weave densities to create customized embroidery effects.

Shape setting: Simplified the cognitive difficulty of the embroidery process by setting different shapes for the nodes on the embroidery path (Figure 5f).

Position setting: Free adjustment of the embroidery node position provides more options to optimize the embroidery pattern.

Hole size setting: users can freely adjust the hole size and the corresponding thread thickness to match different embroidery requirements.

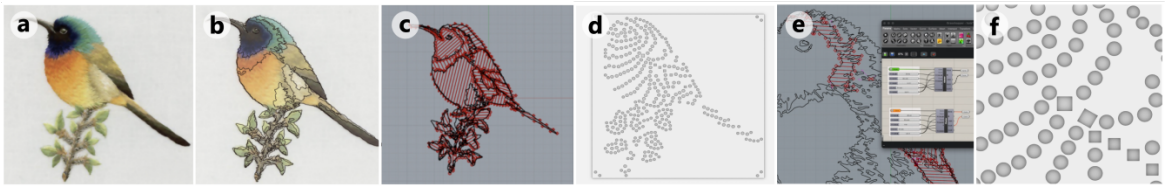


Figure 5: The process of the design tool: users can input a (a) target picture that can be divided into (b) different area based on the colors, then (d-f) customize the parameters to generate the mold file which can be digital fabricated.

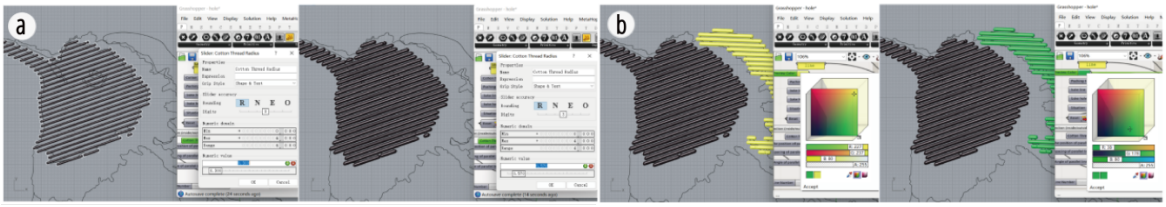


Figure 6: Real-time preview tool in the user interface: the knitting thread's (a) thickness selection; and (b) color selection.

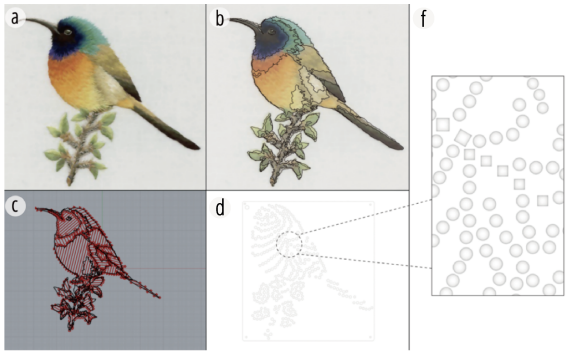


Figure 7: The making process of the bird case (the thickness of the mold is 4mm).

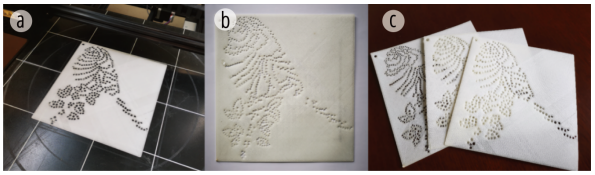


Figure 8: The mold can be printed by a consumer-grade 3D printer.

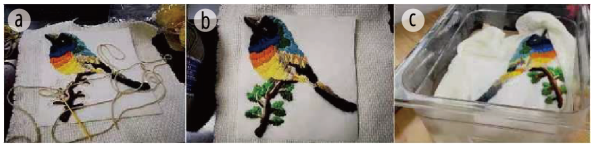


Figure 9: Hand embroidery process with the help of the aided mold (a, b), which can be dissolved in the water (c).

5.5 Real-time preview tool

The tool facilitates an advanced preview, enabling the selection of knitting thread thickness (Figure 6a) and the choice of color for distinct areas (Figure 6b). During the real-time preview, the mold outline and line direction intended for printing are visible, allowing users to modify them promptly based on the image.

6 CASE STUDY IN A HYBRID CRAFT WORKFLOW

We conduct a case study to verify the feasibility of our Embroiderer workflow. This case takes the pattern of a bird resting on a branch as an example and builds a model of the embroidery assist tool for the pattern based on the data such as color distribution, stroke direction, and distribution position of the pattern object in the Rhinoceros. The size of the bird case model is 194mm*184mm (Figure 7). Models can be set up and rendered in the mobile app to preview the final look.

The size of the auxiliary tool is affected by the size of the final knitted product. The mold size is general based on the embroidery pattern. Since the dissolution rate of PVA, a water-soluble material, is related to the thickness, the thicker the thickness, the slower the water dissolution rate. Based on our experiments, the mold's thickness of 0.4mm is optimal.

We determined the size of the mold according to the size of the product to be weaved and prepared 2.5 mm diameter cotton threads of various colors.

In the design and fabrication process, we built a model of the auxiliary embroidery tool based on the given pattern, the determined dimensions of the expected product, and the diameter of the cotton thread, and printed out the model using PVA, a water-soluble material, with a consumer-grade 3D printer (see Figure 8).

After completing the custom mold, we selected a 7cm plastic needle to assist with the embroidery because of its safety and ease of threading and finished the pattern according to the distribution of the holes in the tool (see Figure 9).



Figure 10: The application case of customizing clothes woven with the bird pattern.

Completing the handwork, we put the finished braid into the water to dissolve the braiding aid completely, thus obtaining the final work (see Figure 10).

7 USER STUDY

In order to test the efficiency and usability of using our system, and to investigate the advantages and limitations of hand embroidery creative methods, we recruited participants to learn and practice hand embroidery skills and make freestyle creations. We will introduce and analyze user reactions and feedback, as well as how the Embroiderer system helps the embroiderer work.

7.1 Participants

The participants in the experiment were volunteers, and six undergraduate/graduate students (4 females and 2 males, average age: 20.04 years, ± 0.82 years) were enrolled in the university. They may come from different ages and backgrounds, but they are all interested in participating in embroidery activities and have some relevant foundations. Before the experiment began, participants signed an informed consent form to allow them to participate in the experiment and record their behavior.

7.2 Tools and materials

In the user study, we provide the basic tools and materials to complete the learning and free-style creation procedures.

Needle: In order to perform embroidery, participants were provided with a needle for piercing the fabric and performing embroidery.

Scissors: Scissors are used to trim thread and make other necessary cuts.

Two mm thick silk thread: Participants can use the provided silk thread for embroidery, which may be a standardized type of wire.

Three dimensional printing templates: Participants can use predesigned 3D printing templates that may contain the basic structure of embroidery patterns.

7.3 Procedures

The following is the process of conducting our experiment.

7.3.1 Teaching procedure. The teaching procedure started by demonstrating each step of our method to the participants, showing the previously designed weaving case and how to perform weaving. The purpose of this step is to familiarize participants with the basic methods and techniques of weaving.



Figure 11: The results from participants: (a) P1, (b) P2, (c) P3, and (d) P4.



Figure 12: The results from participants: the custom-made garments embroidery with (a, b) hot air balloons from P5 and (c, d) landscape patterns from P6.

7.3.2 Provide case studies. We provided some woven case studies to the participants. These cases include designs of varying difficulty levels from simple to complex, in order for participants to apply the digital weaving skills they have learned from teaching presentations in practice.

7.3.3 Free choices. After participants become familiar with basic weaving techniques, they will have the opportunity to freely choose or design their own weaving works. Participants do not have specific project selection requirements when participating in the design, they can choose 3D printing templates such as numbers, letters, patterns, etc. provided by us to weave; In addition to the tools and materials we provide, they can freely choose needles, thread colors, fabric materials, etc. They can also start new designs based on their own creativity, which increases creative freedom. The purpose of this stage is to investigate participants' understanding and application abilities of the learned digital weaving techniques, as well as their creativity.

7.3.4 Observation records and data collection. Throughout the entire experimental process, we observed and recorded the behavior and feedback of the participants, including photos of the embroidery projects completed by the participants, time records, notes of design decisions, user satisfaction surveys, feedback opinions, and any other relevant information.

The purpose of this experiment is to study the creativity, decision-making, and preferences of participants in the embroidery process, in order to evaluate how they use the provided tools and materials, and to understand their experience with embroidery activities. These data can be used to improve the design of tools and materials, or to provide insights on user behavior and creative processes for further research and development.

7.4 Results and User feedbacks

7.4.1 Freedom. Participants are satisfied with the freedom of weaving and human-machine interaction systems. P1, P2, P3, P5, and P6

expressed their preference for the personalized adjustment options provided by the system. This allows us to freely adjust the size, arrangement, and style of the patterns, making our embroidery projects more unique. This degree of freedom greatly increases participants' creative motivation, allowing them to better express their creativity. From selecting a simple image in the early stages of the production process to creating excellent embroidery effects based on personal preferences, users can independently decide on the direction and style of creation. This flexibility encourages participants to try different creative methods, thereby improving the overall diversity and uniqueness of their creations.

7.4.2 Accessibility. At the beginning of production, participants started with a relatively simple image and completed embroidery using the PVA material and parameters we provided. Our experiment found that all participants can complete embroidery within half an hour during the learning process of the system. After familiarizing themselves with this process, they chose their favorite pattern, divided paths and colors, and produced excellent pattern effects (Figure 11 and Figure 12). During this process, the participant quickly completed embroidery through the mold, which not only saved time but also achieved a better level of embroidery. All participants unanimously believe that this weaving and human-machine interaction system is very simple and easy to use. P6 added, "Due to the convenient operation process provided by the system, we can focus on creativity without worrying too much about technical details".

7.4.3 Creativity. The results of user study indicate that the system can provide a certain degree of creative freedom, such as P1 said, "I can make personalized adjustments according to my own needs, such as changing the size and arrangement of modes. This simple operation design greatly reduces the threshold of embroidery, making it easy for novice craftsmen to start, P3 Said: "This system allows participants to use the personalized adjustment options provided by the system to freely change the size and arrangement of patterns according to our needs to adapt to different embroidery projects. The feedbacks from participants shows that the entire process has achieved significant results in reducing the threshold of embroidery, which allows participants to creatively design according to actual needs and increases the possibility of creation.

The efficiency of the system has also been highly praised by participants. For example, P4 mentioned "I can easily create custom molds without the need for deep skill advantages. The use process is simple and convenient, as only the submitted patterns need to be woven, and selecting sizes and styles can easily obtain the molds." Compared to traditional manual weaving work, using this system can quickly generate molds, greatly reducing production time. This gives participants more time to try different projects and increases their creative productivity.

Participants' expectations for continuous improvement and innovation in the system indicate their expectations for future functionality and features, hoping to see more exciting developments. These positive user feedback proves that weaving and human-machine interaction systems have broad application prospects in the fields of handicrafts and creativity.

8 LIMITATION, DISCUSSION AND FUTURE WORK

8.1 Technical Requirements

The user experience of our method can be improved. As P4 mentioned that "although the system simplifies the embroidery process in many ways, some users may not feel comfortable with the knowledge and 3D printing technology equipment required to prepare the system." In addition, the richness of custom tools and simulation speed of the software interface can be further improved.

8.2 Limitations on Co-creation

In addition, some participants mentioned that the customization options offered by the system are limited and cannot meet their complex creative needs. They may need to seek more creative freedom with other tools or methods.

These negative reviews highlight the issues we need to consider when developing and promoting woven and human-computer interaction systems. In order to better meet the needs of users, our development team provides more training and support, while constantly improving the design and performance of the system to address potential technical and creative limitations. In addition, we believe that while the barriers to entry for weaving should be lowered, the importance of maintaining traditional weaving skills should be emphasized to ensure that craftsmen can use this system while maintaining their unique skills.

8.3 Dependence

Although participants in user experiments gave positive feedback about our system, the extent to which technology is involved in manual creation still deserves further exploration. P2 expressed concern after the test: "Over-reliance on such a system may weaken the skills of traditional hand-weaving, and I am concerned that craftsmen may lose some of their traditional skills and experience."

8.4 Sustainability

We provide a DIY method to transform household products, which can extend the life cycle of products and contribute to sustainable development; It is also a method of handicraft popularization, which is conducive to the spread of handicraft culture; The materials used in the system are degradable and environmentally friendly.

8.5 Future Work

We hope to continuously address the limitations of the system through long-term optimization. In the future, this method can also have more scalability, materials, modeling (3D space), and application scenarios. For example, users can use more materials, wood, and bamboo. While ensuring the original parameters, the mold has a three-dimensional dimension. The technology can be applied to the packaging of fabric materials, clothing, and other fields. The required 3D embroidery work is produced in the same way through the die holes through the lines conforming to the parameters.

9 CONCLUSION

In this paper, we present Embroiderer as a hybrid craft method, which brings new opportunities and challenges for embroidery technology. Based on material and structural experiments, we developed a computer-aided craft tool to demonstrate the potential of Embroiderer. As a democratized method, Embroiderer hopes to further expand the application potential of embroidery in the field of modern design.

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